

Police Patrol (patrol)

Valerio is robbing a bank, but an alarm started ringing as he was trying to break into the caveau. He needs to get out as fast as possible through the twisted sewer tunnels running nearby the bank.



Figure 1: A sewer tunnel.

The police is already looking for him and has sent K patrols, indexed from 0 to $K - 1$ to guard the manholes connected to the sewer system. There are N manholes, indexed from 0 to $N - 1$ and M tunnels connecting pairs of them. Valerio's friend Filippo is waiting for him nearby manhole $N - 1$, if he can reach it, he will escape safely.

Valerio starts from manhole 0 and, every minute, he can choose to move to a manhole adjacent to the one he is in or to stay another minute under the same manhole. Each police patrol guard L_i manholes, indexed from 0 to $L_i - 1$. Patrol i is initially guarding manhole $H_{i,0}$, every minute it moves from manhole $H_{i,j}$ to manhole $H_{i,j+1}$, after reaching manhole H_{i,L_i-1} , it return to manhole $H_{i,0}$.

 Among the attachments of this task you may find a template file `patrol.*` with a sample incomplete implementation.

Input

The first line contains three integers N , M and K , the numbers of manholes, tunnels and patrols respectively.

The following M lines contain two integers each: the manholes connected by tunnel i .

The following K lines contain the integer L_i followed by L_i integers $H_0, H_1, \dots, H_{L_i-1}$.

Output

You need to write a single line with an integer: the minimum amount of minutes needed to get from

manhole 0 to manhole $N - 1$, or -1 if it is impossible to do.

Constraints

- $2 \leq N \leq 10\,000$.
- $0 \leq M \leq 50\,000$.
- $0 \leq K \leq 100\,000$.
- $1 < L_i \leq 7$ for $i = 0 \dots K - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- Subtask 1 (0 points) Examples.
- Subtask 2 (20 points) $N \leq 100, M \leq 100, K \leq 500$
- Subtask 3 (10 points) $K = 0$
- Subtask 4 (30 points) $L_i \leq 2$ for $i = 0 \dots K - 1$
- Subtask 5 (40 points) No additional limitations.

Examples

3 2 1 0 1 1 2 3 1 2 0	2
5 4 3 0 1 1 2 2 3 3 4 3 4 2 3 2 2 1 2 4 3	5

Explanation

In the **first sample case** Valerio is beneath manhole 0, and there is a single patrol watching manhole 1. In the first minute Valerio moves to manhole 1 while the patrol moves to manhole 2. In the second minute Valerio moves to manhole 2 while the patrol moves to manhole 0, making it out safely in 2 minutes.

In the **second sample case** it takes Valerio 5 seconds to escape while avoiding the patrols.